



Cushing syndrome due to primary bilateral macronodular adrenal hyperplasia

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INTRODUCTION

Among the causes of [Cushing syndrome](#) are three rare types of nodular adrenocortical diseases that are usually bilateral ([table 1](#)):

- [Corticotropin \(ACTH\)-dependent bilateral macronodular adrenal hyperplasia](#), secondary to long-term [adrenal](#) stimulation in patients with [Cushing disease](#) or ectopic [ACTH](#) syndrome. In this etiology, the [adrenals](#) rarely reach the massive size that can be seen with primary bilateral macronodular [adrenal hyperplasia](#), and plasma [ACTH](#) is elevated or normal.
- [ACTH-independent micronodular adrenal hyperplasia](#) and its most frequent variant, primary pigmented nodular adrenocortical disease (PPNAD), which may be sporadic or familial (as part of the [Carney complex](#)); in this disorder, [adrenal](#) micronodules are <1 cm, and often [adrenals](#) are of normal size.

 Primary bilateral macronodular [adrenal hyperplasia](#) (PBMAH), a disorder previously referred to as [ACTH-independent bilateral macronodular adrenal hyperplasia](#) (AIMAH). PBMAH is an **uncommon** cause (<2 percent) of [endogenous Cushing syndrome](#). Treatment of PBMAH with [overt Cushing syndrome](#) is usually surgical. (See '[Treatment](#)' below.)

PBMAH will be reviewed here. Other causes of [Cushing syndrome](#) and PPNAD are discussed separately. (See "[Causes and pathophysiology of Cushing syndrome](#)" and "[Cushing syndrome due to primary pigmented nodular adrenocortical disease](#)".)

TERMINOLOGY

Primary bilateral macronodular adrenal hyperplasia (PBMAH) has been described by various terms, including corticotropin (ACTH)-independent macronodular adrenal hyperplasia (AIMAH), primary macronodular adrenal hyperplasia (PMAH), massive macronodular adrenocortical disease (MMAD), autonomous macronodular adrenal hyperplasia (AMAH), ACTH-independent massive bilateral adrenal disease (AIMBAD), and "giant" or "huge" macronodular adrenal disease. The 2022 World Health Organization pathological classification of adrenal tumors suggests using the term bilateral macronodular adrenal disease [1]. However, we prefer and use the term primary bilateral macronodular adrenal hyperplasia (PBMAH) [2].

EPIDEMIOLOGY

PBMAH is an uncommon cause (<2 percent) of endogenous overt Cushing syndrome, characterized by enlarged adrenal glands containing multiple nonpigmented nodules that are greater than 10 mm in diameter but can be as large as 30 to 40 mm in diameter [3]. It can be diagnosed in patients with Cushing syndrome but more often in those with incidentally found bilateral macronodular adrenals and modest secretion of cortisol who are between the ages of 40 and 60 years. PBMAH is one of the most frequent causes of bilateral adrenal incidentalomas, but its precise prevalence has not been determined in population studies [2,4,5].

In PBMAH, plasma ACTH levels become progressively suppressed when cortisol secretion is sufficiently elevated (increased urinary free cortisol [UFC] levels).

PBMAH is more frequently familial than previously believed (autosomal dominant transmission), and approximately 25 percent of apparent sporadic cases carry a germline mutation of armadillo repeat-containing 5 gene (ARMC5).

PATHOGENESIS

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a aberrant hormone receptors, local production of corticotropin (ACTH) in adrenal tissues, and genetic mutations have been identified in the pathogenesis of PBMAH [2].

Aberrant hormone receptors — The most prevalent abnormality in patients with PBMAH results from the aberrant adrenal expression of ectopic receptors or increased activity of eutopic peptide hormone receptors (table 2) [2,6,7]. Cortisol secretion

In vivo screening protocol to detect the presence of aberrant adrenal hormone receptors in adrenal Cushing's syndrome

Time (min)	Day 1	Day 2	Day 3
-100	Feeding routine	Feeding routine	Feeding routine
-120	+	+	+
0	Baseline ^a	Basal ACTH (mcg/dL)	Basal cortisol (ng/dL)
+100	Baseline ^a	+	+
+120	Baseline ^a	+	+
+140	Baseline ^a	+	+
+160	Baseline ^a	+	+
+180	Baseline ^a	+	+
+200	Baseline ^a	(Peak)	+
+220	Basal insulin ^b		Insulin (mU/L)
+240	+		Insulin (mU/L)
+260	+		+
+280	+		+
+300	+	Basal ACTH (mcg/dL)	+

becomes driven by a hormone that escapes cortisol-mediated feedback.

In a series of patients with PBMAH (and mild or overt Cushing syndrome) who were screened for aberrant receptors, more than 85 percent had at least one and usually several aberrant responses (most commonly to vasopressin) [8-12]. Examples of aberrant receptors include the following:

- **Vasopressin** – Vasopressin-responsive Cushing syndrome is the most frequent aberrant stimulator of cortisol secretion in patients with PBMAH. In these patients, cortisol secretion increases with upright posture or other physiological stimuli of endogenous vasopressin, such as hypertonic saline infusion [6,7,10,13-20].

The V1-vasopressin receptor is normally expressed in the adrenal cortex but is not a major regulator of steroidogenesis. In patients with PBMAH, the exaggerated steroidogenic responses to vasopressin are secondary to the increased activity and/or expression of a "eutopic" receptor-effector system.

- **Beta-adrenergic** – PBMAH and Cushing syndrome have been reported in patients with aberrant adrenal expression of beta-adrenergic receptors. Plasma cortisol and aldosterone levels increase in response to elevations in endogenous catecholamines that occur with upright posture, insulin-induced hypoglycemia, and exercise [21]. Isoproterenol infusion stimulates cortisol and aldosterone secretion, and propranolol blocks this effect and reduces basal cortisol secretion with long-term control of hypercortisolism [10,18,22,23]. These effects were not seen in normal subjects.

The combined presence of adrenal beta-adrenergic receptor and V1-vasopressin receptor has been observed in some patients and families with PBMAH [10,18,21,22]. The combined overexpression of beta-2-adrenoreceptor, serotonin 5-HT4 receptor, and V1-vasopressin receptor has also been reported [23,24].

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• **hCG** – Hypercortisolism due to aberrant luteinizing hormone (LH)/human chorionic gonadotropin (hCG) receptors was first identified in a woman with history of transient Cushing syndrome during sequential pregnancies [25]. In one patient, persistent Cushing syndrome and PBMAH developed after menopause, likely due to higher serum LH concentrations [26]. Administration of a GnRH agonist, leuprolide acetate, resulted in suppression of endogenous LH and normalization of cortisol production [26]. Similar cases have also been reported [25,27,28].

Test	ACTH 1-24 (pg/mL)	CRH (μg)	Vasopressin (pg/mL)
1	+	+	+
2	+	+	+
3	+	+	+
4	+	+	+
5	+	+	+

Notes: + Blood samples for determination of cortisol, ACTH, other hormones, and other agents. * Availability of other components varies between different countries. Not available in some studies; vasopressin can be replaced by terlipressin 1 mg in some countries. † Measurement and interpretation of these results is difficult; refer to Chapter 10 of the text for details.

In vivo screening protocol to detect the presence of aberrant adrenal hormone receptors in adrenal Cushing's syndrome

Table 2 - larger image below

- **Serotonin** – Other cases of aberrant receptors for serotonin, either alone or in combination, have been reported [13,14,29]. In one of the patients described above [26], serotonin 5-HT4 receptor agonists, [cisapride](#) and [metoclopramide](#), also stimulated plasma cortisol. A number of cases with aberrant serotonin receptors have been reported [13,14,20,29-32]. In most cases, a large overexpression of 5-HT4 receptor was found in the adrenals [33-35].
- **GIP** – The ectopic expression of glucose-dependent insulinotropic peptide (also known as gastric inhibitory polypeptide [GIP]) receptor in zona fasciculata cells causes "**food-dependent**" **cortisol production**. Plasma cortisol levels may be relatively low during fasting (low levels of ACTH and GIP) and increase transiently following meals [36-43]. GIP-dependent Cushing syndrome has been reported in approximately 40 patients with PBMAH [44]. *KDM1A* inactivation has been described in patients with GIP-dependent PBMAH. (See '[Genetic causes](#)' below.)

GIP receptors are not expressed in the normal adult or fetal zona fasciculata [38-41,45]. The adrenal nodules in some patients with GIP-dependent PBMAH can progress asynchronously, first in one adrenal, later in the other [41]; in one report, the adrenal adjacent to the macronodules was not atrophic but showed diffuse adrenal hyperplasia with ectopic GIP receptor expression [41], suggesting that larger nodules were the result of secondary clonal proliferation events [46].

In some cases, GIP-dependent PBMAH zones were mixed with bilateral myelolipomas [43]. In a retrospective study of 17 patients with GIP-dependent PBMAH, four had myelolipomas and three had myeloid hyperplasia [47].

- **Angiotensin** – In a patient with PBMAH and a large increase in plasma aldosterone and cortisol levels during upright posture, short-term oral administration of the angiotensin II receptor antagonist [candesartan](#) completely inhibited the elevation in cortisol and aldosterone [48]. In vitro stimulation of cortisol secretion by angiotensin II was also found in patients with PBMAH and Cushing syndrome who had increases in cortisol levels with upright posture [14].

Other receptors – In a study that looked to identify novel abnormally expressed G-protein-coupled receptors (GPCRs) in patients with PBMAH, several new aberrant receptor genes were identified, one of which (the alpha-2A-adrenergic receptor [*ADRA2A*], highly expressed in 13 of 18 BMAHs) may be a potential target for pharmacologic treatment of Cushing syndrome due to PBMAH [49]. [Clonidine](#) (an *ADRA2A* agonist) administration resulted in an increase in cortisol production; the increase could be blocked by an *ADRA2A* antagonist, yohimbine. In other patients, the administration of [glucagon](#) increased plasma cortisol, while [octreotide](#) inhibited plasma cortisol in patients with PBMAH and mild Cushing syndrome [8]. (See '[Screening for aberrant receptors](#)' below.)

Paracrine secretion of ACTH — The hypercortisolism associated with primary PBMAH was previously considered to be corticotropin (ACTH) independent. However, it appears that cortisol secretion in PBMAH is at least partially regulated by intra-adrenal ACTH, which is produced by steroidogenic cells in the hyperplastic adrenals [50,51].

In a report of 30 patients with PBMAH, proopiomelanocortin (POMC) mRNA was detected in 26 PBMAH adrenal gland samples analyzed; moderate/intense immunostaining for ACTH was detectable in clusters of steroidogenic cells in most samples [51]. Conversely, no ACTH staining was present in normal adrenal cortex or cortisol-secreting adenomas. ACTH secretion was confirmed by the demonstration of an ACTH gradient in adrenal vein sampling in two PBMAH patients.

Genetic causes — Several molecular causes have been identified in the pathogenesis of primary PBMAH, indicating that it is a heterogeneous disease [2,9,52]. Most cases of PBMAH were thought to be sporadic [53-56]. However, there are now several reports of familial cases of PBMAH whose presentation suggests autosomal dominant transmission [18-20,57-62].

- **Familial cases** – In familial cases of PBMAH, a number of aberrant hormone receptors, but no genetic abnormality, had been identified in individual families investigated initially, including V1-V2- and V3-vasopressin [19], beta-adrenergic [60], combined V1-vasopressin and beta-adrenergic [18,23], and combined 5-HT4 and V1-V2-vasopressin [20] (see '[Aberrant hormone receptors](#)' above). Several genetic abnormalities have now been identified in PBMAH kindreds.

- **Apparently sporadic cases**

- ***ARMC5* pathogenic variants** – Germline mutations of *ARMC5* are found in approximately 25 percent of apparently sporadic PBMAH patients [52,63-67]. In contrast, germline *ARMC5* mutations are rarely found in patients with bilateral adrenal incidentalomas [68,69]. However, in a recent Italian study, germline *ARMC5* variants were found in 18.8 percent of cases of bilateral incidentalomas but only if those with modest abnormal cortisol secretion; surprisingly, a few cases had *ARMC5* somatic mutations in PBMAH tissues without germline mutations, but this needs further confirmation [70].



ARMC5 behaves as a tumor suppressor gene, and different somatic *ARMC5* mutations or deletions were found in each adrenal nodule examined and appeared to accelerate the proliferation of affected cells [63,71,72]. In two studies, a high percentage of first-degree relatives of the probands carried a germline *ARMC5* mutation and unsuspected PBMAH [63,65].

A high prevalence of biallelic *ARMC* mutations were also identified in other families with PBMAH [23,73-76]. The relationship between *ARMC5* mutations and aberrant receptors or ACTH expression has not been studied in detail yet; no *ARMC5* mutation carriers had GIP-dependent Cushing syndrome, but some had aberrant regulation by vasopressin, beta-adrenergic, and serotonin agonists [23,66,74].

Somatic mutation of *ARMC5* was also found in meningiomas, which can occur in familial PBMAH, suggesting that other tumors could result from mutations of the *ARMC5* tumor suppressor gene (which is expressed in several human tissues) [19,42,65,74].

ARMC5 pathogenic variants are associated with a more severe phenotype in most cases [77,78]. Impact of *ARMC5* mutations is now better understood. *ARMC5* anchors a multiple subunit ubiquitin ligase (E3) which regulates RPB1, the largest subunit of DNA-dependent RNA polymerase II [79]. The loss of *ARMC5* modifies expression of several genes implicated in proliferation and steroidogenesis [79].

- **KDM1A inactivation** – *KDM1A* inactivation has been described in patients with GIP-dependent PBMAH [80]. In a retrospective study of 17 cases, germline mutations and second somatic events in *KDM1A* (lysine demethylase 1A, also known as *LSD1*) gene were identified in all 17 individuals. Mutations were not found in 25 non-GIP-dependent PBMAH nor in GIP-dependent unilateral cortisol secreting adenomas [47]. Functional studies showed that inactivation of KDM1A protein leads to important overexpression of the GIP receptor, but also of other receptors including for LH/hCG, while decreasing others including for vasopressin (V1), serotonin (HT4R), and ACTH (MC2R) [47].

In one family, seven mutation carriers also presented with multiple myeloma, monoclonal gammopathy of undetermined significance, rectal cancer at age 35 years, and one individual had a bronchial neuroendocrine tumor [47].

In a second cohort of 10 women with GIP-dependent PBMAH, hereditary germline *KDM1A* mutations and chromosome 1p second events were identified in nine of the 10 patients (including familial cases), but were not found in 30 non-GIP-dependent PBMAH cases [81]. Thus, GIP-dependent PBMAH with Cushing syndrome can be familial and results from a two-hit inactivation of *KDM1A*, consistent with the tumor suppressor gene model of tumorigenesis.

Penetrance of adrenal disease is low in germline carriers as second somatic event is required, but we offer genetic testing and counseling to these patients and their relatives [2].

- **McCune-Albright syndrome** – Cushing syndrome associated with McCune-Albright syndrome (MAS) is a rare variant of PBMAH that presents in infancy or childhood. MAS is a sporadic disease characterized by polyostotic fibrous dysplasia with café au lait spots, sexual precocity, and other endocrine and nonendocrine disorders [33,82]. An activating somatic of the alpha-subunit stimulatory guanine nucleotide-binding protein, Gs, occurs in some adrenal cells in a mosaic pattern, which leads to the formation of adrenal nodules [83,84]. Activating mutations of Gs alpha have been occasionally found in the PBMAH nodules of pediatric or adult patients without any other features of MAS [84,85]. (See "[Precocious puberty: Diagnosis and evaluation](#)", section on 'McCune-Albright syndrome'.)
- **MEN1** – Bilateral adrenal nodules also have been described as part of multiple endocrine neoplasia syndrome type 1 (MEN1). A retrospective analysis of 715 MEN1 patients found the presence of adrenal lesions in 20 percent, and of those with lesions >1 cm, 12.9 percent had bilateral lesions without further evaluation of how many were PBMAH; most unilateral or bilateral tumors were nonfunctional, but primary aldosteronism was more frequent than in a control group of adrenal incidentalomas without MEN1 [86]. (See "[Multiple endocrine neoplasia type 1: Clinical manifestations and diagnosis](#)".)
- **Familial adenomatous polyposis** – PBMAH or adrenal nodules have been reported in patients with familial adenomatous polyposis and a mutation in the adenomatous polyposis coli (APC) gene [9,87] and with hereditary leiomyomatosis and renal cell cancer disorders due to mutations in the fumarate hydratase (FH) gene [88]. (See "[Clinical manifestations and diagnosis of familial adenomatous polyposis](#)" and "[Hereditary kidney cancer syndromes](#)", section on 'Hereditary leiomyomatosis and renal cell cancer syndrome'.)
- **PDE11A** – Missense germline mutations in phosphodiesterase 11A isoform 4 gene (PDE11A) have been identified initially in kindreds with nonpigmented micronodular hyperplasia but have now been found at higher frequency in patients with various adrenocortical tumors, including PBMAH [89]. As noted above, PBMAH is thus probably genetically heterogeneous, with occasional cases that occur in association with other syndromes [9,34,87,88,90,91], and this has been reviewed in further detail [92-94]. (See "[Cushing syndrome due to primary pigmented nodular adrenocortical disease](#)".)



CLINICAL FEATURES

- **Bilateral adrenal incidentalomas, mild cortisol production** – The majority of patients with PBMAH have only mild cortisol production and present with bilateral adrenal incidentalomas [3]. In these patients, no symptoms of Cushing syndrome may be present, and the nodular

adrenals are identified when abdominal imaging (ultrasound, computed tomography [CT] scan, or magnetic resonance imaging [MRI]) are performed for unrelated reasons. (See ["Evaluation and management of the adrenal incidentaloma"](#).)

In patients in whom cortisol secretion is sufficient to produce mild or overt Cushing syndrome, the symptoms and signs are similar to other etiologies of primary adrenal Cushing syndrome. Plasma ACTH levels become progressively suppressed when cortisol secretion is sufficiently elevated (increased urinary free cortisol [UFC] levels). (See ["Epidemiology and clinical manifestations of Cushing syndrome"](#).)

- **PBMAH with overt Cushing syndrome** – Laboratory findings are similar to those seen in corticotropin (ACTH)-independent Cushing syndrome with unilateral tumors and include the following (see ["Establishing the cause of Cushing syndrome"](#), section on 'ACTH independent (Primary adrenal)'):
 - Increased serum and 24-hour urinary free cortisol (UFC) and low to undetectable plasma ACTH (<10 pg/mL [<2 pmol/L]) in the basal state and after administration of corticotropin-releasing hormone (CRH) [[53-56](#)]. (Cut-offs vary depending on normal range of ACTH assays.)
 - Increased late-night salivary cortisol levels.
 - Serum dehydroepiandrosterone sulfate (DHEAS) levels are usually suppressed as ACTH levels become suppressed; however in some cases, androgen cosecretion occurs, and modestly elevated levels of DHEAS can be present.
 - Cortisol production is not suppressed by low- or high-dose [dexamethasone](#) tests.
 - An exception to this general pattern occurs in patients with gastric inhibitory polypeptide (GIP)-dependent Cushing syndrome in whom cortisol hypersecretion occurs in response to meals and serum cortisol may be low in the fasting state [[36,37](#)]. (See ["Aberrant hormone receptors"](#) above.)



Steroid hormone synthesis is relatively inefficient in PBMAH, as a consequence of decreased steroidogenic enzymatic activity resulting frequently in elevated 17-hydroxyprogesterone levels basally or after stimulation with ACTH [[8,13](#)].

- Co-secretion of serum aldosterone, 18-hydroxycorticosterone, corticosterone, and [estrone](#) may cause hypertension, primary aldosteronism, or feminization in the rare patients in whom they are increased [[28,95,96](#)].

- **Enlarged adrenals** – Primary PBMAH is associated with adrenal glands weighing from 25 to 500 g (normal range 4 to 6 g per adrenal) that contain multiple nonpigmented nodules greater than 10 mm in diameter [53-55]. The enlargement of the adrenal glands is not associated with pain. The nodules appear to be typical benign adrenal nodules, but the internodular cortex may be either hyperplastic (in most cases) or atrophic. PBMAH may present with asynchronous nodular hyperplasia in one gland and later in the contralateral gland [3,41,65].
- **Later age at onset** – Patients with PBMAH and Cushing syndrome typically present in the fifth and sixth decades [53-56], a later age of onset compared with unilateral adrenal cortisol-producing adenomas, primary pigmented nodular adrenocortical disease (PPNAD), or pituitary corticotroph adenomas (Cushing disease).

INITIAL EVALUATION

PBMAH presenting as bilateral incidentalomas

Hormonal hypersecretion — Most patients with PBMAH present with bilateral incidentalomas and mild cortisol excess (see '[Clinical features](#)' above). Their initial evaluation is similar to that for a unilateral incidentaloma and includes tests to look for hormonal hypersecretion, most importantly, the degree of cortisol excess (Cushing syndrome), but also catecholamine and aldosterone excess (eg, pheochromocytomas and aldosteronomas) (see "[Evaluation and management of the adrenal incidentaloma](#)", section on '[Evaluation for hormonal secretion](#)' and "[Evaluation and management of the adrenal incidentaloma](#)", section on '[Mild autonomous cortisol secretion](#)'):

- 1 mg overnight [dexamethasone](#) suppression test
- 24-hour urinary metanephrine or plasma free metanephrine levels
- Aldosterone/renin ratio (only if the patient has hypertension)

In **contrast** with unilateral incidentalomas, the following additional testing is also indicated in patients with bilateral macronodules, including:



Early-morning cortisol and corticotropin (ACTH) levels to rule out primary adrenal insufficiency from bilateral infiltrative/infectious/metastatic diseases.

- A serum 17-hydroxyprogesterone level is measured to rule out nonclassic congenital adrenal hyperplasia (NCAH) due to 21-hydroxylase deficiency, which may sometimes present as silent bilateral adrenal incidentalomas. (See "[Nonclassic congenital adrenal hyperplasia due to 21-hydroxylase deficiency in children and adolescents](#)", section on '[Initial testing \(morning 17-hydroxyprogesterone level\)](#)'.)

As noted, the majority of patients with primary PBMAH have no or mild clinical features of Cushing syndrome. They have normal urinary free cortisol (UFC), detectable ACTH levels still responsive to corticotropin-releasing hormone (CRH), but with subnormal suppression of plasma cortisol following [dexamethasone](#) [6,7,13]. In these patients, low- or high-dose dexamethasone will produce partial suppression of cortisol but not to <1.8 mcg/dL (50 nmol/L) in the morning following nighttime 1 mg dexamethasone administration [6,11,13].

After a nighttime dose of [dexamethasone](#) (1 mg), a range of morning cortisol levels from 1.8 mcg/dL (50 nmol/L) to an intermediate level of 5 mcg/dL (140 nmol/L) to >10 mcg/dL (280 nmol/L) can be seen.

- Patients with serum cortisol concentrations >10 mcg/dL typically have elevated UFC, suppressed ACTH, and progressive clinical signs of Cushing syndrome.
- Those between 1.8 and 10 mcg/dL may have minimal clinical manifestations, but this is probably also affected by various factors, including duration of disease and glucocorticoid receptor sensitivity.

One case of massive PBMAH in a male patient was found to produce large amounts of adrenal androgens (dehydroepiandrosterone [DHEA], dehydroepiandrosterone sulfate [DHEAS]) without any cortisol excess (ACTH not suppressed) or clinical symptoms and was detected only as bilateral incidentaloma [97].

Degree of cortisol excess — So, if the initial screening with 1 mg overnight [dexamethasone](#) does not suppress cortisol to <1.8 mcg/dL (50 nmol/L), we perform the following additional tests to evaluate the degree of cortisol oversecretion:

- 24-hour UFC (**twice**). (See "[Establishing the diagnosis of Cushing syndrome](#)", section on '24-hour urinary cortisol excretion'.)
- Salivary late-night cortisol (**twice**). This is a screening test for Cushing syndrome and is described in detail separately. (See "[Establishing the diagnosis of Cushing syndrome](#)", section on 'Bedtime salivary cortisol'.)
- Fasting ACTH (if not done previously).
- Co-secretion of aldosterone can be seen in PBMAH [98,99]. Therefore, we measure an aldosterone-to-renin ratio to detect mild aldosterone excess, even in the absence of hypertension, when UFC is elevated and ACTH suppressed. (See '[Aberrant hormone receptors](#)' above.)

- Additional blood tests such as DHEAS levels, total [testosterone](#), sex hormone-binding globulin (SHBG), and estradiol may be measured depending upon the clinical findings (eg, virilization in women, hypogonadism or gynecomastia in men, and patients with elevated UFC and suppressed ACTH).

Imaging findings — The results of adrenal computed tomography (CT) and magnetic resonance imaging (MRI) are variable, ranging from apparently unilateral nodules to massive bilateral enlargement [41,53-56,100]. Multiple nodules may be seen that are greater than 10 mm in diameter, but can be as large as 30 to 40 mm in diameter [3] (55 mm in one study [100]). However, in some cases, diffuse enlargement of both adrenals can be present without distinct nodules. In one series of 12 patients, 11 had enlarged multinodular glands; 9 patients underwent pituitary CT or MRI, and all were normal [100]. In PBMAH, in contrast to unilateral adrenal lesions, an unenhanced density above 10 or 20 Hounsfield units are frequently observed but are not associated with malignancy [2].

Adrenal scintigraphy with iodine-131 (131-I)-labeled cholesterol showed bilateral uptake in four of five patients in one report [100]. Although not widely available, this imaging technique could help identify which gland to remove if one chooses unilateral adrenalectomy. A preliminary study indicated increased fluorodeoxyglucose-positron emission tomography (FDG-PET) signal in PBMAH tissues despite absence of malignancy; the frequency of such increased FDG-signal uptake remains to be confirmed in larger cohorts [101,102].

DIAGNOSIS

The diagnosis of PBMAH is made in patients with:

- Overt Cushing syndrome and suppressed corticotropin (ACTH) levels in whom bilaterally enlarged nodular adrenals are found at abdominal computed tomography (CT) scan (a minority of cases).

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Incidentally found bilateral adrenal hyperplasia with several macronodules in whom a subnormal suppression of cortisol is found (>1.8 mcg/dL or >50 nmol/L) following 1 mg overnight [dexamethasone](#) suppression test (the majority of cases).

However, in individuals with single nodules in each adrenal and subclinical cortisol secretion, the diagnosis could be bilateral adenomas or an early phase of PBMAH.

Differential diagnosis — Primary PBMAH, an ACTH-independent form of Cushing syndrome, must be distinguished from ([table 1](#)):

Frequency of causes of Cushing syndrome

Diagnosis	Percent of patients
ACTH-dependent Cushing syndrome	80
Cushing disease	68

- Unilateral cortisol-producing adrenal adenomas
- Primary pigmented nodular adrenocortical disease (PPNAD)
- Secondary ACTH-dependent macronodular adrenal hyperplasia (Cushing disease or ectopic ACTH secretion)

Features that can help distinguish between these disorders include the following:

- Primary PBMAH versus secondary ACTH-dependent macronodular hyperplasias – Measure plasma ACTH:
 - Plasma ACTH is suppressed by excess cortisol in PBMAH (<10 pg/mL [<2 pmol/L]). In contrast, ACTH is inappropriately normal or slightly high in ACTH-dependent macronodular hyperplasia (≥ 15 pg/mL [3.3 pmol/L]). (See "[Establishing the cause of Cushing syndrome](#)".)
 - Rare patients with ACTH-dependent hyperplasia have low but consistently detectable plasma ACTH concentrations (≤ 10 pg/mL [2 pmol/L]), presumably due to its suppression by some of their adrenal nodules, which have acquired the capacity to secrete cortisol constitutively by unidentified mechanisms [103,104]. (See "[Establishing the cause of Cushing syndrome](#)".)
- Among the different primary adrenal sources of Cushing syndrome, there are several features that are suggestive of PBMAH:
 - Most patients with overt Cushing syndrome due to PBMAH present in the fifth and sixth decades [53-56], a later age of onset compared with unilateral cortisol-producing adenomas or PPNAD.
 - Size of adrenal gland nodules usually clearly distinguish PPNAD (micronodular) from PBMAH (macronodular).



While the typical laboratory findings of PBMAH with overt Cushing syndrome include high fasting serum and urinary cortisol and suppressed plasma ACTH and dehydroepiandrosterone sulfate (DHEAS) concentrations [53-56], an exception to this general pattern occurs in patients with PBMAH and gastric inhibitory polypeptide (GIP)-dependent Cushing syndrome, in whom cortisol hypersecretion occurs in response to meals [36,37].

Ectopic ACTH syndrome	12
Ectopic CRH syndrome	$<<1$
ACTH-independent Cushing syndrome	20
Adrenal adenoma	10
Adrenal carcinoma	8
Micronodular hyperplasia	<1
Macronodular hyperplasia	<1
Pseudo-Cushing syndrome	
Major depressive disorder	1
Alcohol use disorder	$<<1$

Frequency of causes of Cushing syndrome

Table 1 - larger image below

- **Metastatic or infiltrative adrenal diseases** – In PBMAH, the increase in cortisol following administration of exogenous ACTH is greater than in normal adrenals and reflects the large adrenal cell mass [8,10]. This biochemical finding can be useful in distinguishing PBMAH from other causes of bilaterally enlarged adrenals such as metastatic or infiltrative diseases, where steroidogenic response to ACTH administration is decreased. In PPNAH, cortisol response is usually limited following exogenous ACTH stimulation.

FURTHER EVALUATION AFTER DIAGNOSIS

Further evaluation is needed in patients diagnosed with PBMAH and overt or mild Cushing syndrome.

Metabolic consequences of hypercortisolism — Once the diagnosis of PBMAH is established, further evaluation is necessary to look for potential metabolic consequences of cortisol excess. Patients should undergo evaluation for glucose intolerance and type 2 diabetes (fasting glucose, glycated hemoglobin [A1C] levels, or oral glucose tolerance test [OGTT]), hypertension (including 24-hour monitoring if necessary), coronary heart disease, and bone health (bone mineral density and vertebral fracture assessment).

Screening for aberrant receptors — In our center and several other academic centers [3,8-11,13,16-22,24,26,27,29-32,36,38-42,105], we suggest that patients with PBMAH and clinical or mild Cushing syndrome undergo screening for aberrant receptors as this may change the therapeutic strategy ([table 2](#)). Protocols have been developed to systematically evaluate for these aberrant receptors ([table 2](#)) [6-11,14].

One group suggests restricting tests to those for which a therapy is possible, ie, mixed meal, posture test, and luteinizing hormone-releasing hormone (LHRH) test [106]. However, ideally, patients should be referred to expert centers that offer testing for aberrant receptors.

The strategy is to examine whether cortisol or other steroids change in response to physiological (fasting, right posture, mixed meals) and pharmacologic (gonadotropin-releasing hormone [GnRH], [opressin](#), [glucagon](#), [metoclopramide](#)) tests that modulate the levels of ligands for the potential aberrant receptors. Of note, thyrotropin-releasing hormone (TRH) testing has rarely detected aberrant responses, and TRH is not widely available. We therefore do **not** consider TRH testing to be necessary when screening for aberrant receptors.

Screening family members — First-degree relatives (>25 years of age) of patients with PBMAH who do not carry armadillo repeat-containing 5 (*ARMC5*) gene mutations should be screened for PBMAH by first looking for evidence of hypercortisolism. This can be done with an overnight 1

mg [dexamethasone](#) test; those who do not suppress their plasma cortisol on the following morning below 1.8 mcg/dL (50 nmol/L) should undergo adrenal computed tomography (CT) scan to look for possible features of PBMAH (adrenal nodules, bilateral enlargement). (See ["Dexamethasone suppression tests"](#), section on 'Overnight screening test' and 'Imaging findings' above.)

- Family members who are found to have PBMAH should undergo detailed biochemical and clinical investigation described above and targeted evaluation for the aberrant hormone receptors that were found in the initial case if it can lead to specific targeted medical therapy.
- Patients with PBMAH associated with other genetic causes such as multiple endocrine neoplasia syndrome type 1 (MEN1) or familial adenomatous polyposis should undergo the recommended evaluation/screening for those disorders. (See ["Multiple endocrine neoplasia type 1: Clinical manifestations and diagnosis"](#), section on 'Monitoring for MEN1-associated tumors' and ["Familial adenomatous polyposis: Screening and management of patients and families"](#).)
- The search for germline *ARMC5* mutations (and possibly other genes as more are identified) now allows familial screening through a simple blood test and identify who should undergo further evaluations [50,63]. (See ["Genetic causes"](#) above.)
- The author of this topic suggests that all patients with PBMAH be tested for ectopic GIP receptor because *KDM1A* pathogenic variants would likely be present. Those with the pathogenic variant are at high risk for familial cases and associated malignancies [107].

Testing can be performed by measuring serum cortisol before and two hours after an oral 75 g glucose load or a mixed meal. A more than 50 percent reproducible increase in cortisol would warrant genetic testing for *KDM1A* mutation [2].

TREATMENT

 Treatment of PBMAH with overt Cushing syndrome is usually surgical; bilateral adrenalectomy is previously utilized in most patients with PBMAH and severe Cushing syndrome (UFC >3 times upper limit of normal), but more recently, unilateral adrenalectomy was found to normalize UFC in a high proportion of patients with less severe Cushing syndrome [2,108].

In patients in whom aberrant hormone receptors are identified, specific pharmacologic therapies have been utilized in selected cases as alternatives to adrenalectomy. (See ["Pharmacologic therapy"](#) below.)

The initial patients identified with PBMAH were identified because of severe Cushing syndrome and they underwent bilateral adrenalectomy. However, considering the wide range of clinical presentations from asymptomatic bilateral incidentaloma cases to overt Cushing syndrome, we currently recommend the following approaches based on severity of cortisol excess.

Patients without aberrant receptors

Overt Cushing syndrome

Bilateral adrenalectomy

- **Candidates** – We suggest bilateral surgical adrenalectomy for patients with PBMAH, equally enlarged macronodular adrenals, and clinically severe Cushing syndrome (elevated urinary free cortisol [UFC] >3 times above upper limit of normal in repeated samples) who have **no** evidence of aberrant receptors that would be amenable to specific medical therapy [109].

Surgical bilateral adrenalectomy is uniformly effective in these patients. Unilateral adrenalectomy is sometimes performed if one adrenal is much larger than the other as this could also result in normalization of UFC without permanent adrenal insufficiency [108,110]. Surgical adrenalectomy, including surgical approach and postoperative hormone replacement, is discussed in greater detail separately. (See "[Persistent or recurrent Cushing disease: Surgical bilateral adrenalectomy](#)" and "[Adrenalectomy techniques](#)".)

Adrenal enzyme inhibitors — Adrenal enzyme inhibitors are sometimes given to patients with overt Cushing syndrome to control cortisol secretion before surgery, particularly if surgery needs to be delayed for some underlying condition such as infection, recent cerebrovascular or coronary event, or pulmonary emboli [111-113]. (See "[Medical therapy of hypercortisolism \(Cushing syndrome\)](#)", section on 'Adrenal steroidogenesis inhibitors'.)

- **Post-adrenalectomy** – Post-adrenalectomy, these patients will require lifetime glucocorticoid and mineralocorticoid replacement but are **not** at risk for Nelson syndrome (corticotroph tumor progression), because the pituitary is intrinsically normal. Therefore, pituitary radiation to prevent Nelson syndrome is unnecessary. (See "[Persistent or recurrent Cushing disease: Surgical bilateral adrenalectomy](#)", section on 'Monitor for Nelson syndrome'.)
- **Thromboprophylaxis** – We agree with the Endocrine Society guidelines and suggest thromboprophylaxis in patients with overt Cushing syndrome going to surgery because of



their high risk of thromboembolism [114]. (See "[Epidemiology and clinical manifestations of Cushing syndrome](#)", section on '[Thromboembolic events](#)').)

Modest cortisol excess

Unilateral adrenalectomy — In patients with moderately increased cortisol production (the majority of PBMAH patients with Cushing syndrome have less than threefold increase in UFC levels) but with clinical evidence of cortisol excess, we suggest unilateral adrenalectomy as it can restore UFC levels to normal [108,115-118]. However, as the cell mass subsequently increases in the contralateral adrenal, medical therapy with steroidogenesis inhibitors or a second adrenalectomy may become necessary [110,115-117,119,120].

In a review of 286 PBMAH cases from 30 publications treated with unilateral adrenalectomy, initial remission was achieved in 77 percent. 29 percent experienced a recurrence of hypercortisolism, and 31 percent required a completion contralateral adrenalectomy within a median time of 72 months [2]. Transient hypocortisolism occurred in 16 percent and require careful monitoring and replacement with glucocorticoids.

In 15 patients with PBMAH treated by unilateral adrenalectomy, resection of the larger gland led to remission of hypercortisolism after three months in all 15 patients and a low risk of recurrence. In a second study of 25 PBMAH cases undergoing unilateral adrenalectomy, biochemical remission was initially achieved in 21 (84 percent) [119]. After a median of 50 months of follow-up, only eight patients (32 percent) were still biochemically controlled. Adrenalectomy of the contralateral side had to be performed in three (12 percent).

We do not suggest adrenal venous sampling to select which adrenal to remove, as standardization for data interpretation are lacking. In the largest study of 10 patients, an adrenal/peripheral vein ratio of ≥ 6.5 was considered consistent with a cortisol hypersecretion, and a cortisol lateralization ratio of ≤ 2.0 was in favor of bilateral source of cortisol [121]. The largest adrenal gland on computed tomography (CT) scan is selected for unilateral adrenalectomy [2].

Some experts have proposed alternative conservative surgical procedures in PBMAH patients with unilateral adrenalectomy on one side and subtotal contralateral adrenal resection [122,123]. In 17 patients, including 10 patients with modest cortisol excess, 95 percent achieved complete hypercortisolism control postoperatively, while 71 percent experienced recovery of their hypothalamic-pituitary-adrenal (HPA) axis at a median follow-up of 41 months. Only one patient developed relapse of hypercortisolism after 30 months requiring a completion surgery [123]. However, as many patients had only modest cortisol excess, the risk-benefit ratio of performing a bilateral procedure over unilateral

adrenalectomy alone is not clear. It may be further evaluated prospectively in more severe cases rather than bilateral adrenalectomy [2].

Minimal cortisol excess

Monitoring without surgery — For patients with mild cortisol excess (normal UFC, normal corticotropin [ACTH], cortisol suppression following 1 mg overnight [dexamethasone](#) <5 mcg/dL) we suggest monitoring without surgery. PBMAH is a benign process that has not been shown to become malignant. Monitoring should include clinical evaluation and biochemical assessment (1 mg overnight dexamethasone test, UFC, basal ACTH, fasting glucose) every 6 to 12 months, and a yearly CT scan.

We suggest unilateral adrenalectomy if there is disease progression with increased UFC, complete suppression of ACTH, and clinical effects of hypercortisolism such as osteoporosis, diabetes, hypertension, or neuropsychological manifestations (see '[Unilateral adrenalectomy](#)' above). As studies in unilateral adrenal incidentaloma have reported progressive adverse cardiovascular outcomes in patients with plasma cortisol >5 mcg/dL (138 nmol/L) after a 1 mg [dexamethasone](#) test [5,124,125], we are more commonly referring patients to unilateral adrenalectomy when signs of cortisol excess develop.

In a retrospective cohort study of 1048 patients with adrenal incidentalomas, higher post-1 mg [dexamethasone](#) serum cortisol levels were associated with increased mortality [126].

Patients with PBMAH and aberrant receptors

Pharmacologic therapy — In some patients in whom aberrant hormone receptors have been identified, specific pharmacologic therapies blocking the aberrant receptors can be effective as alternatives to adrenalectomy. Despite normalization of cortisol secretion with specific aberrant receptor-targeted therapy, no tumor regression has been observed; this may be secondary to incomplete receptor blockade or to the possibility that proliferation is regulated by armadillo repeat-containing 5 gene (*ARMC5*) mutations or other secondary genetic events [7,46,63,127].



Beta-adrenergic receptors – We suggest beta blocker therapy for PBMAH patients with Cushing syndrome and aberrant beta-adrenergic receptors. In catecholamine-dependent Cushing syndrome and PBMAH, beta-adrenergic receptor antagonists ([propranolol](#)) are effective in the long-term control of hypercortisolism [12,21,24]. Propranolol can be started at 60 mg/day, increasing every two weeks up to 320 mg/day in two to three divided doses in order to achieve mid-normal levels of UFC. We suggest monitoring patients weekly initially

for the first month (UFC, blood pressure, pulse), monthly for three months and every three months if stable afterwards.

In patients who experience side effects such as nightmares or who have compliance problems with split doses of [propranolol](#), we have used [nadolol](#), another nonselective antagonist that is administered as a single daily dose (40 to 320 mg).

Symptomatic bradycardia without reaching normalized UFC may limit the use of beta-adrenergic receptor antagonists so that bilateral adrenalectomy is required. If beta-adrenergic receptor antagonists do not normalize UFC, we suggest adrenalectomy. (See '[Bilateral adrenalectomy](#)' above.)

- **LH or hCG receptors** – We suggest gonadotropin-releasing hormone (GnRH) agonist therapy for patients with luteinizing hormone (LH) or human chorionic gonadotropin (hCG) receptors. In LH/hCG-dependent PBMAH and Cushing syndrome, suppression of endogenous LH levels with long-acting [leuprolide](#) acetate controlled steroid secretion and avoided bilateral adrenalectomy [26,27]. We typically use a monthly dose of depot leuprolide acetate (3.75 mg intramuscular).

Levels of 24-hour UFC, morning plasma cortisol, and LH levels are monitored two and four weeks after the first injection. Treatment is continued if UFC normalizes or improves at least by 50 percent within the first three months based upon two UFC samples each month. If UFC levels are not normalized within three months, unilateral or bilateral adrenalectomy is recommended, depending on levels of cortisol excess as described previously.

For patients who are well controlled on GnRH agonist therapy, we suggest follow-up every three to four months for the first year (with UFC, morning plasma cortisol, and serum LH). Of note, we have followed patients every six months thereafter; some have remained controlled for more than 10 years. In premenopausal women or males, we suggest adding gonadal steroids to the GnRH agonist to avoid the consequences of hypogonadism (estrogen-progestin therapy in women and [testosterone](#) in men). We also suggest monitoring bone mineral density. (See "[Endometriosis: Long-term treatment with gonadotropin-releasing hormone agonists and antagonists](#)", section on '[GnRH with add-back therapy](#)'.)

GIP-dependent Cushing syndrome – In patients with gastric inhibitory polypeptide (GIP)-dependent Cushing syndrome, we suggest surgical adrenalectomy rather than pharmacologic therapy (see '[Aberrant hormone receptors](#)' above). Pharmacologic blockade of postprandial release of GIP with [octreotide](#) or [pasireotide](#) leads to clinical and biochemical improvement of Cushing syndrome, but the beneficial effect does not persist in the long term, probably because of eventual desensitization of somatostatin receptors in GIP-

secreting duodenal K cells [37,38,42]. Thus, we suggest surgical adrenalectomy in these patients.

- **Other** – When specific receptor antagonists for ACTH, [vasopressin](#), serotonin, GIP, or other aberrant receptors become available, a broader range of pharmacologic therapies could become useful [50,51]. Until these become available, surgical adrenalectomy is recommended.

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Diagnosis and treatment of Cushing syndrome](#)".)

SUMMARY AND RECOMMENDATIONS

- **PBMAH** – Primary bilateral macronodular adrenal hyperplasia (PBMAH) has been described by various terms, including corticotropin (ACTH)-independent macronodular adrenal hyperplasia (AIMAH), primary macronodular adrenal hyperplasia (PMAH), massive macronodular adrenocortical disease (MMAD), autonomous macronodular adrenal hyperplasia (AMAH), ACTH-independent massive bilateral adrenal disease (AIMBAD), and "giant," "huge," macronodular adrenal disease, or bilateral macronodular adrenal disease. (See '[Terminology](#)' above.)
- **Epidemiology** – PBMAH is more frequently familial than previously believed (autosomal dominant transmission), and approximately 25 percent of apparent sporadic cases carry a germline mutation of armadillo repeat-containing 5 gene (*ARMC5*). (See '[Epidemiology](#)' above.)
- **Pathogenesis** – PBMAH results in hypercortisolism through a number of mechanisms that are described above. In the majority of patients with PBMAH, cortisol appears to be regulated by a number of aberrant G-protein-coupled receptors (GPCRs) and by locally produced ACTH. (See '[Pathogenesis](#)' above.)
- Aberrant receptors include [vasopressin](#), serotonin 5-hydroxytryptamine 4 (5-HT4), gastric inhibitory polypeptide (GIP), beta-adrenergic, luteinizing hormone (LH)/human chorionic gonadotropin (hCG), and others. (See '[Aberrant hormone receptors](#)' above.)
- Inactivating biallelic inactivation of *KDM1A* (lysine demethylase 1A) is responsible for glucose-dependent insulinotropic polypeptide (GIP)-dependent PBMAH and is associated with

increased risk of multiple myelomas and other malignancies. PBMAH appears heterogeneous as several other genetic abnormalities can be rarely associated. (See '[Genetic causes](#)' above.)

- **Clinical features** – The clinical presentation and biochemical findings of Cushing syndrome in PBMAH are similar to those seen in other types of ACTH-independent Cushing syndrome. However, patients with PBMAH typically have a later age of onset (fifth and sixth decades), and patients with GIP-dependent Cushing syndrome have a unique pattern of cortisol hypersecretion ("food dependent") ([table 1](#)). (See '[Clinical features](#)' above.)
- **Diagnosis** – The diagnosis of PBMAH is made in patients with overt Cushing syndrome and suppressed ACTH and dehydroepiandrosterone sulfate (DHEAS) levels with bilaterally enlarged nodular adrenals. The diagnosis is most frequently made in patients with incidentally found bilateral diffusely enlarged adrenals or with multinodular adrenals and mild hypercortisolism in whom a subnormal suppression of cortisol is found (>1.8 mcg/dL or >50 nmol/L) following 1 mg overnight [dexamethasone](#) suppression test. (See '[Diagnosis](#)' above.)
- **Further evaluation after diagnosis** – Patients with PBMAH and clinical or mild Cushing syndrome should undergo testing for aberrant receptors as this may change the therapeutic strategy ([table 2](#)). (See '[Aberrant hormone receptors](#)' above and '[Screening for aberrant receptors](#)' above.)

We test first-degree relatives of index cases without known genetic mutation using 1 mg overnight [dexamethasone](#) suppression test for the presence of familial PBMAH.

Genetic screening for *ARMC5* (or eventually other genes responsible for PBMAH) has become available to screen first-degree relatives of index cases. The author of this topic suggests screening for ectopic GIP receptor in PBMAH patients as it would warrant screening for *KDM1A* mutation, familial screening, and surveillance for associated malignancies. (See '[Screening family members](#)' above.)

Management



Patients without aberrant receptors – In patients with PBMAH in whom aberrant hormone receptors are not present or for which no specific pharmacologic blockade is available or effective (see '[Patients without aberrant receptors](#)' above):

- **Severe hypercortisolism** – We perform bilateral adrenalectomy for very severe Cushing syndrome with urinary free cortisol (UFC) >3 times the upper limit of normal and equally enlarged adrenals. (See '[Overt Cushing syndrome](#)' above and '[Bilateral adrenalectomy](#)' above.)

In similar patients but with very asymmetric adrenal size, unilateral resection of the largest adrenal can be performed. (See '[Unilateral adrenalectomy](#)' above.)

Adrenal enzyme inhibitors are sometimes given to patients with overt Cushing syndrome to control cortisol secretion before surgery, particularly if surgery needs to be delayed for some underlying condition such as infection, recent cerebrovascular or coronary event, or pulmonary emboli. (See '[Adrenal enzyme inhibitors](#)' above.)

- **Moderate hypercortisolism** – In patients with moderately increased cortisol production (less than threefold increase in UFC levels) and clinical evidence of cortisol excess, we suggest unilateral adrenalectomy as it can restore UFC levels to normal (**Grade 2C**). Such patients can present transient secondary adrenal insufficiency and require glucocorticoid replacement postoperatively. (See '[Modest cortisol excess](#)' above and '[Unilateral adrenalectomy](#)' above.)
- **Minimal cortisol excess** – In patients with PBMAH and normal levels of urinary cortisol, the decision between observation and surgery should be based on biochemical (suppressed ACTH, elevated late-night salivary cortisol, and cortisol >5 mcg/dL following 1 mg [dexamethasone](#) overnight test) and clinical evidence of cortisol excess (hypertension, diabetes, osteoporosis, or neuropsychological manifestations). (See '[Minimal cortisol excess](#)' above and '[Monitoring without surgery](#)' above.)
- **Patients with aberrant receptors** – Our approach to patients with PBMAH and overt Cushing syndrome regulated by aberrant receptors includes (see '[Patients with PBMAH and aberrant receptors](#)' above):
 - Beta-blockers for patients with aberrant beta-adrenergic receptors. (See '[Pharmacologic therapy](#)' above.)
 - Gonadotropin-releasing hormone (GnRH) agonists (and sex steroid replacement) for patients with LH/hCG receptors. (See '[Pharmacologic therapy](#)' above.)



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